

Competitive Exam Practice 11: Parallel Processing and Pipeline Hazards

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

September 5, 2026

Each question carries 1–2 marks. MCQ = single correct option; MSQ = multiple-select; NAT = Numerical Answer Type. Provenance tags separate verified real past-year questions from original practice material.

- Q1** [*GATE CSE 2002 Q2.6, verified real PYQ — GATEOverflow V3 p.81*] [**MCQ, 1 mark**] The performance of a pipelined processor suffers if:
- (A) the pipeline stages have different delays
 - (B) consecutive instructions are dependent on each other
 - (C) the pipeline stages share hardware resources
 - (D) All of the above
- Q2** [*GATE CSE 2008 Q76, verified real PYQ — GATEOverflow V3 p.83*] [**MCQ, 1 mark**] Delayed branching can help in the handling of control hazards. For all delayed conditional branch instructions, irrespective of whether the condition evaluates to true or false,
- (A) the instruction following the conditional branch instruction in memory is executed
 - (B) the first instruction in the fall-through path is executed
 - (C) the first instruction in the taken path is executed
 - (D) the branch takes longer to execute than any other instruction
- Q3** [*Original, GATE-pattern*] [**NAT, 2 marks**] Six independent instructions execute in an ideal balanced five-stage pipeline with one cycle per stage and no stalls. What are the pipelined completion time and the finite-stream speedup over a non-pipelined execution, respectively? Report the speedup to two decimal places.
- Q4** [*Original, GATE-pattern*] [**NAT, 2 marks**] Consider a five-stage pipeline (IF, ID, EX, MEM, WB) and the sequence `LW R1,0(R2)` followed immediately by `ADD R3,R1,R4` and then `SUB R5,R3,R6`. Load data is available after MEM; ALU results are forwarded at the end of EX. Starting the first IF in cycle 1, in which cycle does `SUB` complete WB, and how many bubbles are required? Give the two integers in the form “cycle, bubbles”.
- Q5** [*Original, GATE-pattern*] [**MCQ, 1 mark**] One vector instruction applies the same addition operation to eight data elements, while four independent cores execute different matrix tiles. Which pair correctly identifies the parallelism in the two cases?

- (A) SIMD, then MIMD
- (B) MIMD, then SIMD
- (C) SISD, then MISD
- (D) ILP, then DLP

Q6 [*Original, GATE-pattern*] [**MCQ, 1 mark**] In a five-stage pipeline, an instruction in MEM and a younger instruction in IF request the same single-ported unified memory in the same cycle. Which is the most direct classification and one valid remedy?

- (A) Data hazard; forward the producer's ALU result
- (B) Structural hazard; split instruction and data memories
- (C) Control hazard; rename the destination register
- (D) Data hazard; insert a branch prediction entry

Q7 [*Original, GATE-pattern*] [**NAT, 1 mark**] Branches comprise 20% of instructions, the branch misprediction rate is 5%, and a misprediction flush costs 3 cycles. Ignoring all other effects, what is the branch contribution to CPI? Report to two decimal places.

Q8 [*Original, GATE-pattern*] [**MSQ, 1 mark**] Which statements are correct for the five-stage model used in this week?

- (A) Forwarding can remove many ALU-to-ALU RAW stalls.
- (B) Forwarding always removes the immediately following load-use stall.
- (C) A branch misprediction can require flushing younger instructions.
- (D) A true RAW dependence can be eliminated by register renaming.